EECS 245, Fall 2026

LEC 13 Null Space and Inverses

→ Read: Ch. 5.3-5.4, 6.2

Agenda

- Brief recap: column space, row space, rank } 5.3
- Null space } 5.4
- Rank-nullity theorem
  → See video playlist on website!
- If time: basics of invertibility

Announcements

- MT 1 regrades due today
- HW 5 due Sunday
- Can follow these notes live → click link on website

Recap: column space, row space, rank

$$A = \begin{bmatrix} 3 & 6 & 0 & 9 & 3 \\ 2 & 4 & 0 & 6 & 2 \\ 0 & 0 & 1 & -5 & 0 \\ 1 & 2 & 0 & 3 & 1 \end{bmatrix}$$

$\vec{a}^{(1)}$ $2\vec{a}^{(1)}$ $\vec{a}^{(3)}$ $\vec{a}^{(4)}$ $\vec{a}^{(5)}$

$3\vec{a}^{(1)} - 5\vec{a}^{(3)}$

Tasks:

1. rank(A) = 2
2. colsp(A) is a 2-dimensional subspace of $\mathbb{R}^4$
3. $\text{colsp}(A^T)$ (row space) is a 2-dimensional subspace of $\mathbb{R}^5$
4. Find two different bases for colsp(A)

4. one basis for colsp(A): $\{\vec{a}^{(1)}, \vec{a}^{(3)}\}$

another basis: $\{\vec{a}^{(4)}, \vec{a}^{(5)}\}$

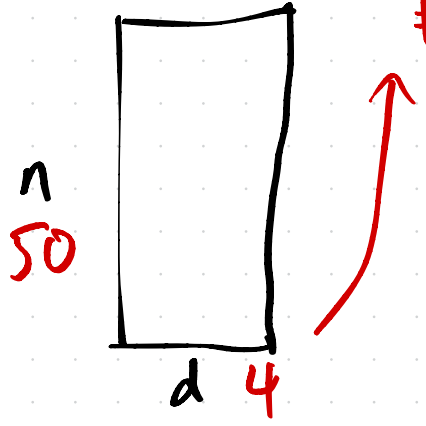
not basis: $\vec{a}^{(1)}, \vec{a}^{(3)}$

Important:

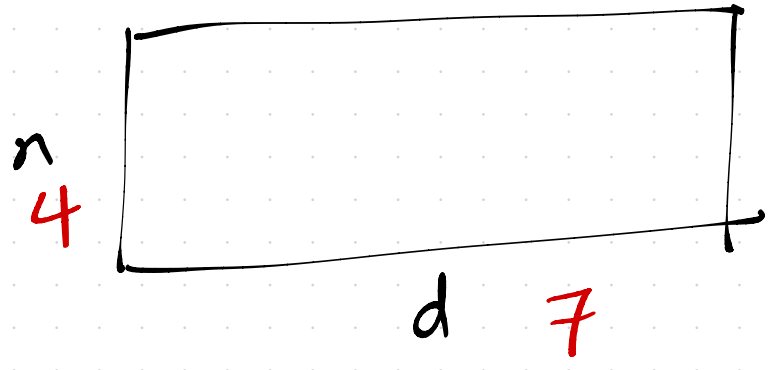
$$\text{rank}(A) = \text{rank}(A^T),$$

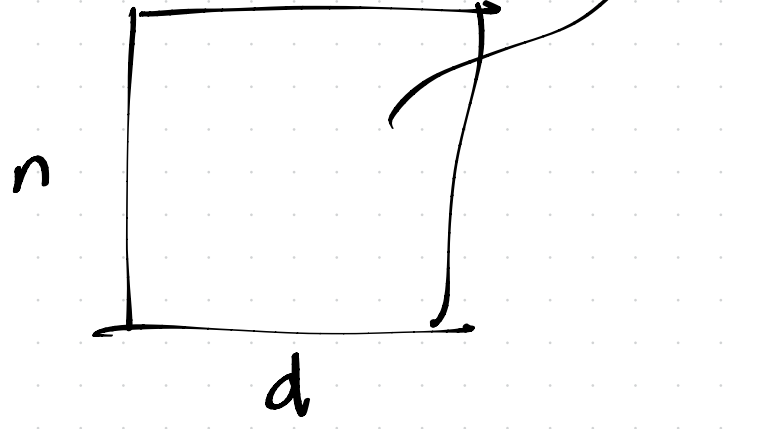
meaning

$$\dim(\text{colsp}(A)) = \dim(\text{colsp}(A^T))$$



linearly independent cols = 4,
so # linearly independent rows = 4,
so rows can't all be independent!





If square ($n = d$),
then if all cols
are linearly independent,
then all rows must
be as well, and
vice versa!

From lab yesterday

$$A\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

$\vec{0} = 3(\text{col } 1) + 2(\text{col } 2)$,

So cols are not linearly independent!

10:50 AM Thu Oct 15 Safari
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Note that the last page of this worksheet is empty i

Activity 5: Can We Work Backward?

Let $A = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix}$.

cols are scalar multiples!

$$A = \begin{bmatrix} 3 & 6 & 0 & 9 & 3 \\ 2 & 4 & 0 & 6 & 2 \\ 0 & 0 & 1 & -5 & 0 \\ 1 & 2 & 0 & 3 & 1 \end{bmatrix}$$

col 1 col 2 col 3 col 4 col 5

col 1 − col 5 = 0

col 5 = col 1

col 2 = 2 col 1

$2\,\text{col}\,1 - 1\,\text{col}\,2 = 0$

col 4 = 3(col 1) − 5(col 3)

$\vec{0} = 3(\text{col}\,1) - 5(\text{col}\,3) - \text{col}\,4$

$\vec{x}$

Goal: find as many linearly independent vectors $\vec{x}$ as possible where

$$A\vec{x} = \vec{0}$$

$$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ -1 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 3 \\ 0 \\ -5 \\ -1 \\ 0 \end{bmatrix}$$

null space

$A : n \times d$

$$\text{nullsp}(A) = \{ \vec{x} \mid A\vec{x} = \vec{0} \}$$

subspace of $\mathbb{R}^d$

Watch the video posted on website about why null space is a subspace

$\vec{x} \rightarrow$ [multiply by A] $\rightarrow \vec{0}$ vector gone!

Example:

$$A = \begin{bmatrix} 1 & 7 \\ 2 & 4 \\ 3 & -1 \end{bmatrix}$$

A's cols are linearly independent, so the only vector in nullsp(A) is $\vec{0}$

$\text{rank}(A) = 2$

$\text{nullsp}(A) = \{ \vec{0} \} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ → "trivial null space"

$\dim(\text{nullsp}(A)) = 0$

col 1 col 2 col 3 col 4 col 5

$$A = \begin{bmatrix} 3 & 6 & 0 & 9 & 3 \\ 2 & 4 & 0 & 6 & 2 \\ 0 & 0 & 1 & -5 & 0 \\ 1 & 2 & 0 & 3 & 1 \end{bmatrix} \qquad \begin{bmatrix} 34 \\ 6 \\ 0 \\ 0 \\ -34 \end{bmatrix}$$

$$A\vec{x} = \vec{0}$$

$$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} 3 \\ 0 \\ -5 \\ -1 \\ 0 \end{bmatrix}$$

all linearly independent

→ all LCs of these 3 $\vec{x}$'s are also in null space

$$\dim(\text{nullsp}(A)) = 3$$

Rank-nullity theorem

For any $n \times d$ matrix A,

$$\text{rank}(A) + \dim(\text{nullsp}(A)) = d$$

d: # of columns

Intuition:

independent + # dependent = # total

Example

"outer product"

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{2\times 1} \underbrace{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}}_{1\times 3}$$

$\text{rank}(B) = 1$

$\dim(\text{nullsp}(B)) = 2$

$B\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \vec{0}$ $B\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \vec{0}$

Find one basis for nullsp(B)

basis : $\left\{ \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

$\underbrace{\text{rank}(B^T)}_{=1} + \underbrace{\dim(\text{nullsp}(B^T))}_{=1} = 2$

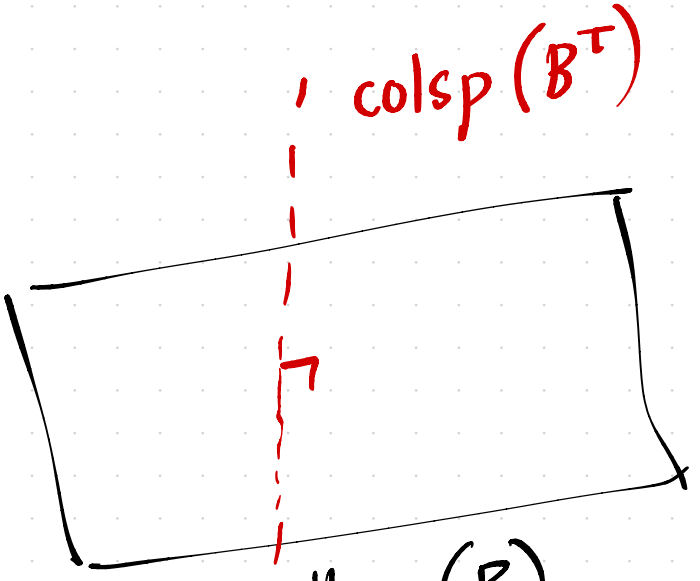
colsp(B) is a line in $\mathbb{R}^2$,

colsp(B^T) is a line in $\mathbb{R}^3$,

nullsp(B) is a plane in $\mathbb{R}^3$

nullsp(B^T) is a line in $\mathbb{R}^2$.

"left nullspace"



$\vec{v}$ must be orthogonal to every row of B!

$$\{ \vec{v} \mid B\vec{v} = \vec{0} \}$$

Video: row space and null space are orthogonal complements

"left null space"

$$B^T \vec{x} = \vec{0}$$

$$\left(B^T \vec{x}\right)^T = \vec{0}^T$$

$$\vec{x}^T B = \vec{0}^T$$

because $\vec{x}^T$ appears in the left

Find a matrix A such that

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} \in \text{colsp}(A)$$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \in \text{nullsp}(A)$$

A : 2×2